

Experimental results: expanded measurement coverage

30 September 2026

This results package covers twenty existing trials. The online proposal must be reconciled with its current source before publication.

1 Experimental results

1.1 Measurement coverage and interpretation

These comparisons report each run separately. Messages generated during evaluation form a distinct broker-acknowledged cohort followed through the original drain cutoff. Completion is the end of application processing before offset-commit acknowledgment. Warm-up messages are excluded from cohort latency and deadline outcomes but can remain queued and can contribute to completion throughput during evaluation. Every ten-minute evaluation is preceded by one minute of warm-up and followed by two minutes of drain: thirteen minutes in total. The earlier five-minute evaluations last eight minutes in total and are reported in the appendix.

The main deadline is one second; the half-second calculation is retained as sensitivity analysis in the appendix. Both were specified before the four-partition trials. Their application to older trials is retrospective; the original 99-ms outcomes remain in the records. These are experimental completion deadlines, not a validated application SLA, and no acceptable violation percentage has yet been defined.

Monitoring summaries and time-series figures cover evaluation, while requested-resource and observed-process integrals cover evaluation plus drain. Actual CPU use and resident memory are distinct from Kubernetes resource requests. Invalid measurements, ownership changes and offset discontinuities are excluded from continuous intervals; no missing period is filled with zero. Individual runs are retained because two repetitions do not establish a general effect or a confidence interval.

The cohort replay retained the original admission, completion and unfinished counts and p99 values. New handover validations passed, and no duplicate completion attempts were found in the ten-minute blocks. These checks support record-level accounting in the synthetic application; they do not establish exactly-once business effects at an external durable sink. Latency uses producer and consumer wall-clock timestamps, while application-processing duration uses a local monotonic timer. Clock synchronization does not imply zero timing uncertainty.

The combined diagnostic figures show persistent-hot counts, lag skew and maximum/mean partition lag for each run. A partition is hot when its lag is greater than 10 offsets and greater than the current mean plus one population standard deviation. It is persistently hot when it is hot now and was hot in at least twelve of fifteen contiguous valid observations. Fifteen points on the two-second grid span twenty-eight seconds between endpoints. Invalid observations, gaps greater than three seconds, ownership changes and decreasing offsets restart the history. Undefined detector values remain missing rather than being shown as zero. These are exploratory diagnostic settings; the experiments use scheduled actions and do not validate adaptive trigger thresholds.

The runtime-state panels reconstruct $\mathcal{X}(t) = (\overline{B}_W(t), \overline{G}_W(t), \overline{S}_W(t), H_{\text{persistent}}(t))$ from the recorded observations. Backlog and skew average fifteen samples, spanning 28 seconds; growth averages fifteen two-second intervals, spanning 30 seconds and requiring sixteen samples. These are the sample and interval conventions in the Metrics formulas. The fourth component is a set: the figure shows its size, while the supporting CSV and JSON retain the exact partition identifiers at each timestamp. A missing set is different from an observed empty set. These states are reconstructed offline; no adaptive controller using them was evaluated. The performance figure separately uses a ten-second processing-backlog growth window (five intervals and six observations). Shortening that plot window does not change the runtime-state or persistence window.

The measured processing backlog uses the contiguous completion frontier, while consumer-position lag uses the position after records have been returned to the application. Their equality at the retained valid snapshots in these two ten-minute blocks does not make the definitions interchangeable, particularly for future batched or concurrent processing. The acknowledged-unfinished curves use message identities and include warm-up work, so they are not the evaluation-cohort unfinished percentage. The metric-by-metric coverage summary is provided in Table 25 in Appendix B.

1.2 Twelve high-input partitions: ten-minute evaluation

The twelve-partition comparison used 700 aggregate messages/s over 60 partitions, with 80% of traffic directed to twelve partitions initially owned by Consumer 2. Eight trials compared retaining the same three-consumer assignment, redistribution within three, scaling to six with targeted redistribution, and scaling to six with Kafka’s normal cooperative rebalance. Each condition was run twice, reversing condition order in the second repetition. Starting ownership and the original consumers’ pod identities, nodes and resource declarations were checked before traffic.

Every trial admitted 420,000 evaluation messages. Keeping the initial assignment left 34.64–34.83% unfinished at cutoff; all interventions completed the cohort. Redistribution within three produced p99 latency of 80.10–84.16 seconds, compared with 100.81–107.24 seconds for targeted scaling and 71.41–72.46 seconds for normal Kafka scaling. The one-second cohort deadline-miss rates reveal a different aspect of the latency distribution: 30.50–31.84% for redistribution, 26.26–27.33% for targeted scaling and 21.05–25.83% for normal scaling, versus 83.32–83.36% without intervention. Thus a higher p99 does not imply more misses at every deadline.

The resource measurements show the trade-off. Redistribution used approximately 1,035–1,038 observed consumer CPU core-seconds over evaluation plus drain; targeted scaling used 1,310–1,327 and normal scaling 1,452–1,459. Requested CPU was 72 core-minutes for redistribution and approximately 137 for either six-consumer response. Actual RSS integrals were 2.53–2.65, 3.95–4.15 and 3.66–3.69 GiB-minutes respectively. These values concern consumer processes and resource declarations, not total infrastructure cost. The first unchanged baseline lacks almost all local request history and cannot support a full-window request-cost comparison; its message outcomes and recovered historical process measurements remain usable within their stated coverage.

Targeted scaling confirmed recovery sooner than redistribution within three (169.0–184.9 versus 305.2–328.9 seconds after the action decision), despite its higher whole-cohort p99. Recovery requires processing backlog at most 100 offsets for thirty consecutive valid seconds while input continues. Normal scaling’s recovery varied from 143.0 to 259.0 seconds. Explicit ownership changes use a global release/acquire/resume barrier; their 14.40–14.54-second and 19.05–19.50-second processing pauses are distinguished from complete action duration. Normal cooperative scaling had much shorter global gaps, which does not imply every partition processed continuously.

Persistent hotspots were present before intervention. In the redistribution and targeted-scaling runs, none remained at any defined snapshot in the final two evaluation minutes. The diagnostic figure also shows that relative skew may rise during successful backlog clearance: the maximum and mean absolute partition lags are then small. Persistence and skew are useful descriptions of concentration, but neither alone establishes overload.

Processing-start mean latency ranged from 7.552 to 160.117 seconds across this block; it remains separate from completion latency. Every trial achieved 700 acknowledged evaluation messages/s. No duplicate completion attempts were detected, so completed-attempt and distinct-completion throughput coincide in these trials. Mean application work is reported below; it excludes evidence-writing and offset-commit overhead. Observed per-process CPU coverage over evaluation plus drain ranged from 87.50% to 99.72%. Coverage uses the complete observation window, including time before additional consumers started; a lower value for a newly started consumer is not automatically a monitoring failure. The integral includes only observed valid process intervals.

Response	Run	p99 (s)	Unfinished (%)	Misses > 1 s (%)	Useful throughput (msg/s)
Keep 3	1	366.64	34.641	83.32	423.23
Keep 3	2	362.79	34.828	83.36	422.62
Redistribute 3	1	80.10	0.000	30.50	727.71
Redistribute 3	2	84.16	0.000	31.84	729.53
Targeted scale 6	1	100.81	0.000	26.26	727.75
Targeted scale 6	2	107.24	0.000	27.33	729.31
Kafka scale 6	1	71.41	0.000	25.83	727.65
Kafka scale 6	2	72.46	0.000	21.05	728.48

Table 1: Whole evaluation-born cohort through the fixed drain cutoff. Deadline misses include overdue unfinished messages. Useful throughput counts distinct completions during evaluation, including warm-up records completed then; it can exceed the input rate while queued work is cleared.

Response	Run	Actual CPU (core-s)	Actual RSS (GiB-min)	Requested CPU (core-min)	Requested memory (GiB-min)
Keep 3	1	892.9	3.24	—	—
Keep 3	2	889.7	3.24	72.00	36.00
Redistribute 3	1	1,035.3	2.53	72.00	36.00
Redistribute 3	2	1,037.9	2.65	72.00	36.00
Targeted scale 6	1	1,310.5	4.15	136.86	68.43
Targeted scale 6	2	1,327.5	3.95	136.80	68.40
Kafka scale 6	1	1,451.5	3.66	137.15	68.58
Kafka scale 6	2	1,459.0	3.69	137.15	68.57

Table 2: Consumer resources over evaluation plus drain. Actual values integrate observed process CPU and RSS intervals and sum them across consumers. Requested values integrate Kubernetes resource declarations; neither is monetary or whole-cluster cost. A dash denotes unavailable full-window request coverage, not zero cost.

Response	Run	Mean total lag	Peak total lag	Lag coverage (%)	Peak partition lag	Mean skew
Keep 3	1	98,929	180,743	99.7	14,752	4.90
Keep 3	2	100,033	182,438	99.7	14,794	4.90
Redistribute 3	1	7,837	46,048	96.3	5,536	9.92
Redistribute 3	2	8,664	47,675	96.3	5,883	10.30
Targeted scale 6	1	8,470	55,923	95.3	5,216	9.05
Targeted scale 6	2	9,520	60,387	95.3	5,544	9.31
Kafka scale 6	1	6,688	38,516	98.0	4,332	10.34
Kafka scale 6	2	5,702	40,065	98.0	4,510	8.91

Table 3: Lag diagnostics during evaluation only, in offsets except skew and coverage. Means use trapezoidal area divided by covered time; peaks are sampled maxima. The time mean of the skew ratio is not the ratio of time-averaged maximum and mean lag. Invalid samples and ownership changes are not bridged.

Response	Run	Mean (s)	p50 (s)	p95 (s)	Mean work (ms)	Observed misses > 1 s (%)
Keep 3	1	158.994	161.677	341.123	2.613	72.41
Keep 3	2	160.120	164.185	344.752	2.614	72.45
Redistribute 3	1	11.585	0.099	62.417	1.956	33.16
Redistribute 3	2	12.676	0.100	65.276	1.943	34.61
Targeted scale 6	1	13.813	0.124	81.024	2.451	29.08
Targeted scale 6	2	15.245	0.125	87.387	2.457	30.27
Kafka scale 6	1	9.140	0.120	57.805	2.764	28.66
Kafka scale 6	2	7.555	0.120	57.333	2.773	24.15

Table 4: Supplementary latency diagnostics. Mean, p50 and p95 use completed evaluation-born messages. Mean work is the local monotonic application-processing duration. Observed violations use completion attempts occurring during evaluation and have a different denominator from the primary admitted-cohort deadline rate.

Response	Run	Decision to active (s)	Global gap (s)	Extra pending during pause	Net pending during transfer	Recovery (s)
Keep 3	1	—	N/A	N/A	N/A	N/A
Keep 3	2	—	N/A	N/A	N/A	N/A
Redistribute 3	1	25.73	14.544	10,181	6,808	305.2
Redistribute 3	2	26.01	14.400	10,081	6,581	328.9
Targeted scale 6	1	72.81	19.498	13,648	2,082	169.0
Targeted scale 6	2	77.18	19.049	13,350	548	184.9
Kafka scale 6	1	31.01	0.092	N/A	-6,777	259.0
Kafka scale 6	2	33.01	0.202	N/A	-9,823	143.0

Table 5: Intervention costs. For targeted actions, transfer is release request to verified active ownership; for native scaling it is decision to stable-six-owner confirmation. The latter also defines native decision-to-active. Recovery is confirmation after the actual decision. Global gap measures the longest interval without any consumer processing, not a per-partition downtime. Negative net pending change means processing reduced outstanding work during the measured transition. Dashes denote unavailable or inapplicable values.

Response	Run	C0	C1	C2	C3	C4	C5
Keep 3	1	0	0	12	—	—	—
Keep 3	2	0	0	12	—	—	—
Redistribute 3	1	4	4	4	—	—	—
Redistribute 3	2	4	4	4	—	—	—
Targeted scale 6	1	2	2	2	2	2	2
Targeted scale 6	2	2	2	2	2	2	2
Kafka scale 6	1	0	0	2	4	2	4
Kafka scale 6	2	0	0	2	4	3	3

Table 6: Final observed numbers of configured high-input partitions per consumer. These are traffic targets, not persistent-hot detector counts. All high-input partitions initially belonged to Consumer 2. A dash means no observed partition ownership.

Response	Run	Reference event	C3 (s)	C4 (s)	C5 (s)
Targeted scale 6	1	Replica request	63.37	63.41	63.37
Targeted scale 6	2	Replica request	67.20	67.12	67.23
Kafka scale 6	1	Action decision	9.22	16.43	14.35
Kafka scale 6	2	Action decision	9.42	18.66	9.54

Table 7: Activation delay to the first valid completion on each added consumer. Targeted scaling uses the recorded replica-request event; native scaling uses the action decision because an equivalent replica-request timestamp was not separately recorded. These are not whole-action completion times or pure rebalance delays.

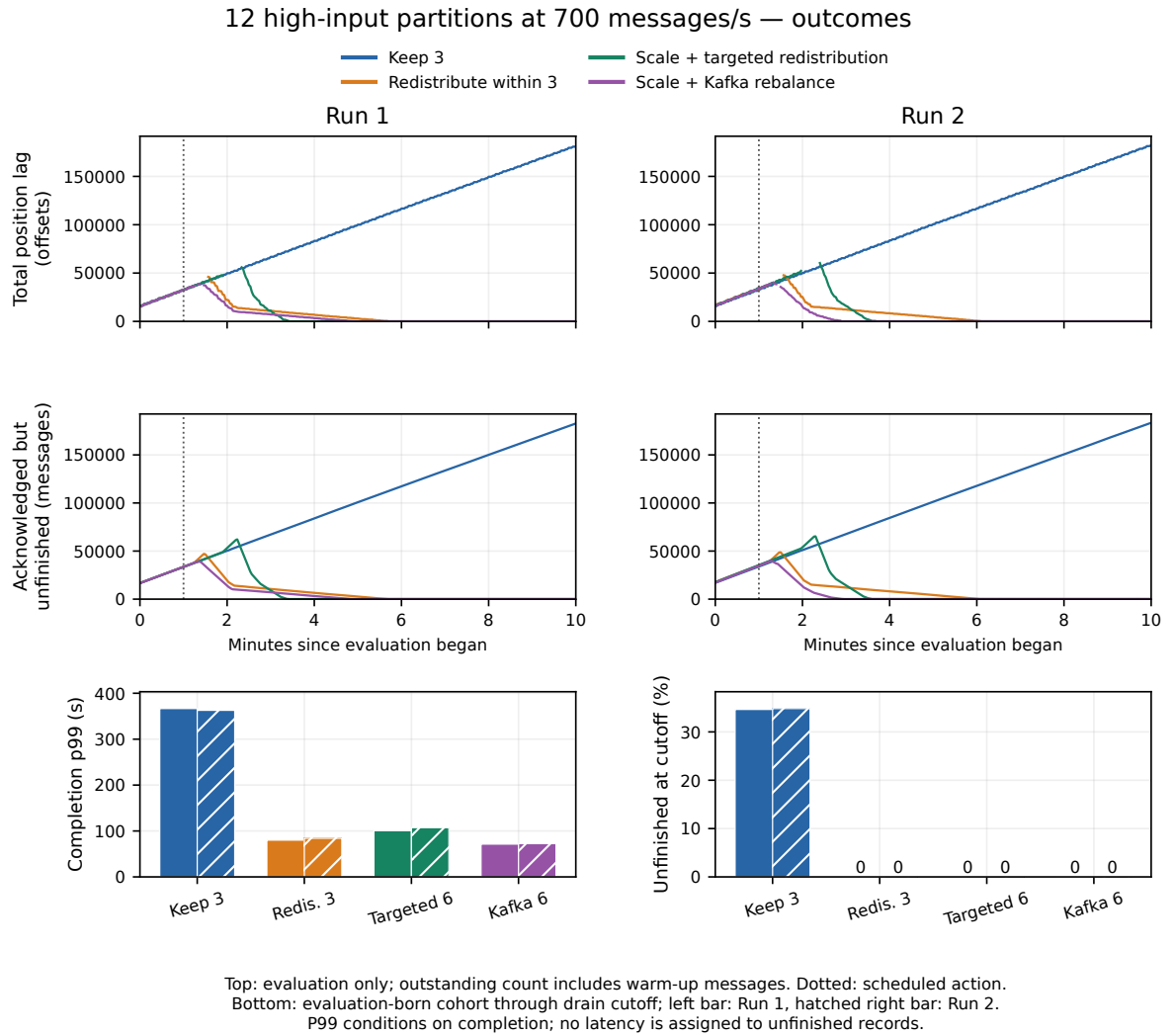
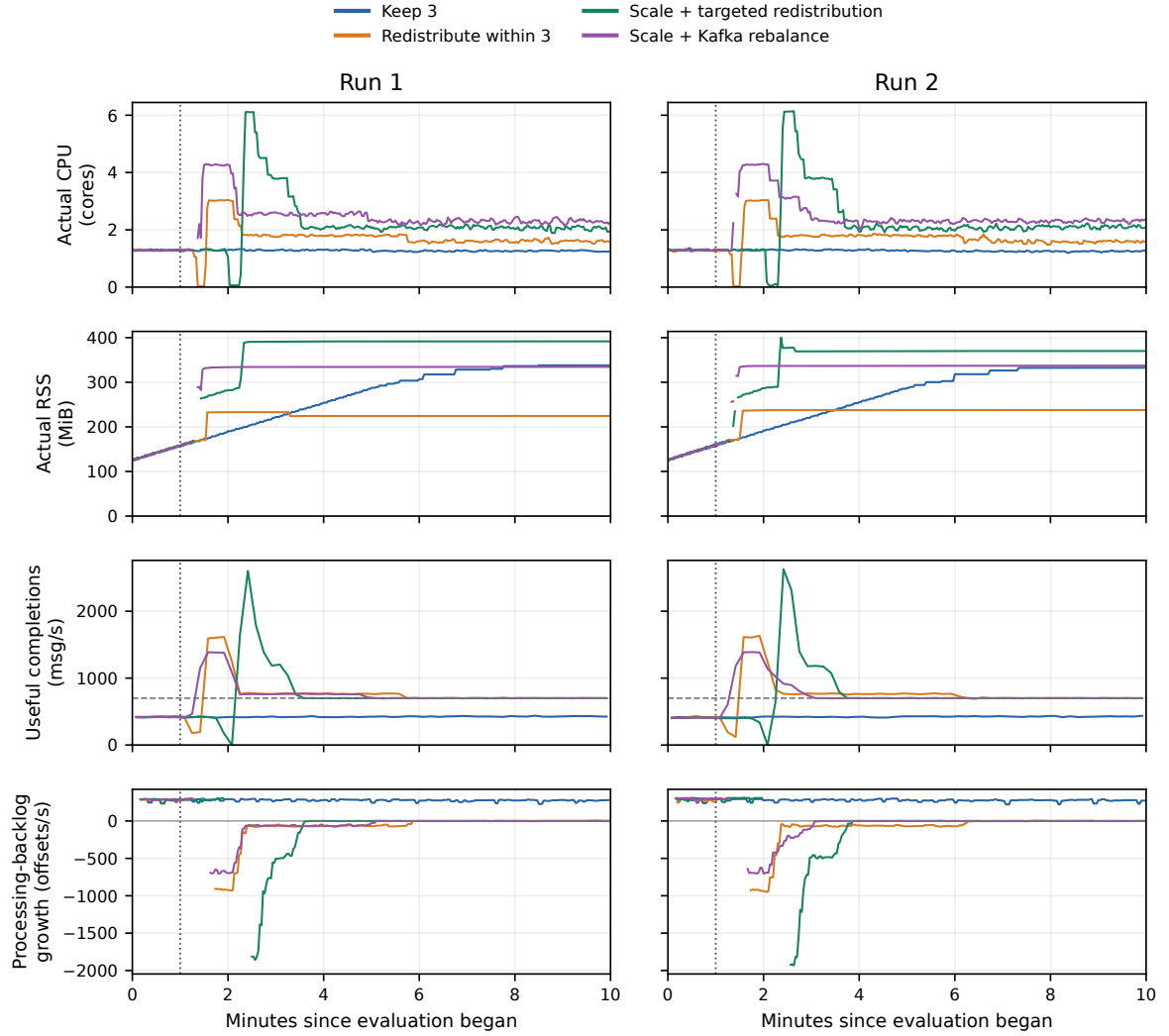


Figure 1: Compact outcome comparison: total consumer-position lag and exact acknowledged-but-unfinished work during evaluation, followed by whole-cohort p99 and unfinished percentage at the drain cutoff. Warm-up work is included in the time-series counts but excluded from the outcome cohort.

12 high-input partitions at 700 messages/s — performance and resources



CPU and RSS sum freshly observed active consumer processes. Unknown observations remain gaps.
Throughput: distinct completions in 10-s bins; dashed: 700-msg/s input. Growth: 10-s window.
Evaluation only; warm-up records completed here count toward throughput. Dotted: scheduled action.

Figure 2: Actual consumer CPU and resident memory, useful completion throughput, and ten-second processing-backlog growth. Resource traces sum fresh measurements for all started consumer processes; an unavailable component leaves a gap. Requested resources and their integrals remain separate in the resource table.

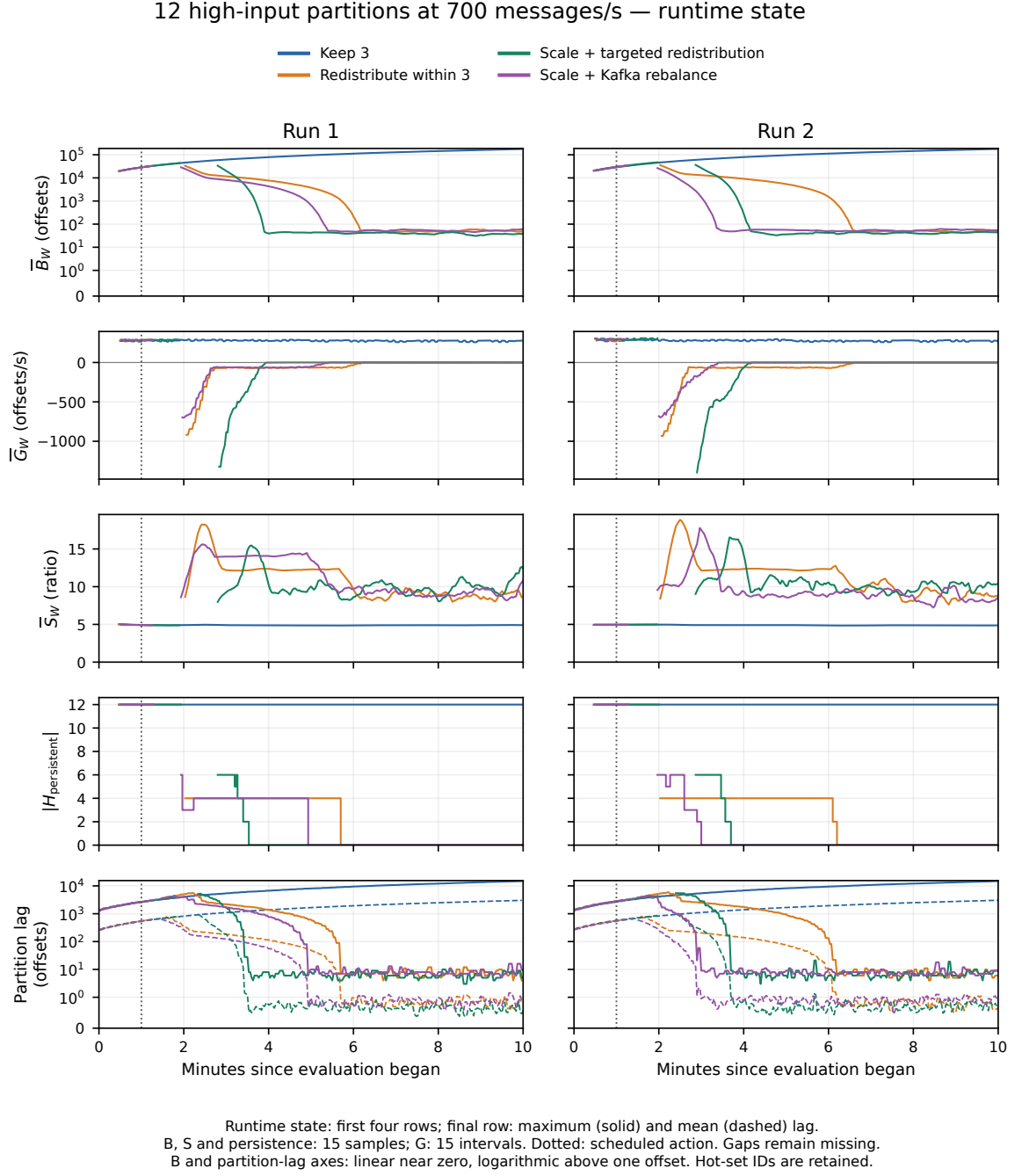


Figure 3: Reconstructed runtime state and partition-lag diagnostics. The first four rows show window-mean total lag, window growth, window-mean skew and persistent-hot count. The final row shows maximum and mean partition lag. Exact persistent-hot identifiers remain in the supporting data; the count alone does not identify a reassignment target. No scheduled action was selected by this reconstructed state.

1.3 Four high-input partitions: ten-minute evaluation

Four further trials retained the same aggregate rate of 700 messages/s, which had shown bounded lag under balanced input during calibration, but directed 80% of input to four of the sixty partitions. All four initially belonged to Consumer 2. Two trials redistributed within three consumers and two scaled to six with targeted redistribution. With three consumers, the target hot-partition counts were 2–1–1; with six they were 1–1–1–1–0–0. Matching initial ownership and original consumer placement was verified, and the second repetition reversed condition order. These assignments concern high-input partitions; they do not measure service capacity or implement a capacity-optimal assignment.

All four trials admitted and completed 420,000 evaluation messages by the fixed cutoff. Redistribution within three had lower p99 in both runs (84.50 and 79.28 seconds) than targeted scaling (110.19 and 112.94 seconds), and fewer one-second deadline misses (26.63% and 25.82%, versus 35.23% and 34.72%). Its mean valid total lag was approximately 6,994–7,296 offsets, compared with 12,683–12,740 for targeted scaling. The sampled peaks were 50,157–50,879 and 66,598–70,217 offsets respectively. Both useful-throughput values and the full time traces are retained so that average completion rate does not hide the transition.

Redistribution used 72 requested CPU core-minutes and 36 requested memory GiB-minutes, whereas targeted scaling used approximately 136.84 core-minutes and 68.42 GiB-minutes over evaluation plus drain. Observed consumer CPU use was approximately 1,007–1,019 versus 1,301–1,302 core-seconds; observed RSS integrals were 2.72–2.75 versus 4.01–4.09 GiB-minutes. Additional requested capacity therefore had both a measured utilization footprint and an allocation cost in these trials.

Redistribution confirmed recovery 173.45–180.85 seconds after the decision, compared with 266.80–290.54 seconds for targeted scaling. Decision-to-active intervals were 21.92–27.26 versus 78.52–81.26 seconds, while processing pauses were 12.31–15.08 versus 19.88–21.73 seconds. Net additional unfinished work during release-to-active handover and additional work accumulated during the processing pause are reported separately because their endpoints differ. These observations support the sufficiency of redistribution within three for the tested workload; they do not establish that scaling cannot help under other conditions.

The prespecified final-two-minute production cohort completed in every trial. Its p99 was 0.187 and 0.349 seconds for redistribution, versus 2.714 and 0.481 seconds for targeted scaling. The first targeted-scaling run experienced a later backlog spike after satisfying the recovery criterion. First recovery confirmation therefore does not prove continuously low backlog thereafter. The earlier twelve-partition late window was selected retrospectively and is not interchangeable with this prespecified analysis.

Unlike the twelve-partition block, small persistent-hot sets sometimes remained in the late four-partition observations. For example, redistribution Run 1 classified a partition with 11 offsets of lag as persistently hot when total lag was only 42. A relative hotspot can therefore persist in a pipeline with low absolute backlog. This motivates interpreting persistence jointly with absolute lag, growth and completion outcomes.

Processing-start mean latency ranged from 9.933 to 20.020 seconds across this block; it remains separate from completion latency. Every trial achieved 700 acknowledged evaluation messages/s. No duplicate completion attempts were detected, so completed-attempt and distinct-completion throughput coincide in these trials. Mean application work is reported below; it excludes evidence-writing and offset-commit overhead. Observed per-process CPU coverage over evaluation plus drain ranged from 87.22% to 99.72%. Coverage uses the complete observation window, including time before additional consumers started; a lower value for a newly started consumer is not automatically a monitoring failure. The integral includes only observed valid process intervals.

Response	Run	p99 (s)	Unfinished (%)	Misses > 1 s (%)	Useful throughput (msg/s)
Redistribute 3	1	84.50	0.000	26.63	730.87
Redistribute 3	2	79.28	0.000	25.82	731.38
Targeted scale 6	1	110.19	0.000	35.23	730.89
Targeted scale 6	2	112.94	0.000	34.72	731.13

Table 8: Whole evaluation-born cohort through the fixed drain cutoff. Deadline misses include overdue unfinished messages. Useful throughput counts distinct completions during evaluation, including warm-up records completed then; it can exceed the input rate while queued work is cleared.

Response	Run	Actual CPU (core-s)	Actual RSS (GiB-min)	Requested CPU (core-min)	Requested memory (GiB-min)
Redistribute 3	1	1,006.9	2.75	72.00	36.00
Redistribute 3	2	1,018.7	2.72	72.00	36.00
Targeted scale 6	1	1,301.6	4.09	136.83	68.42
Targeted scale 6	2	1,300.6	4.01	136.85	68.42

Table 9: Consumer resources over evaluation plus drain. Actual values integrate observed process CPU and RSS intervals and sum them across consumers. Requested values integrate Kubernetes resource declarations; neither is monetary or whole-cluster cost. A dash denotes unavailable full-window request coverage, not zero cost.

Response	Run	Mean total lag	Peak total lag	Lag coverage (%)	Peak partition lag	Mean skew
Redistribute 3	1	7,296	50,879	96.3	13,509	20.56
Redistribute 3	2	6,994	50,157	97.0	12,258	20.88
Targeted scale 6	1	12,683	66,598	95.7	15,839	23.00
Targeted scale 6	2	12,740	70,217	95.0	16,277	22.72

Table 10: Lag diagnostics during evaluation only, in offsets except skew and coverage. Means use trapezoidal area divided by covered time; peaks are sampled maxima. The time mean of the skew ratio is not the ratio of time-averaged maximum and mean lag. Invalid samples and ownership changes are not bridged.

Response	Run	Mean (s)	p50 (s)	p95 (s)	Mean work (ms)	Observed misses > 1 s (%)
Redistribute 3	1	10.749	0.096	67.562	1.874	29.75
Redistribute 3	2	9.935	0.096	65.012	1.886	29.01
Targeted scale 6	1	19.330	0.129	92.980	2.378	37.98
Targeted scale 6	2	20.023	0.127	95.434	2.418	37.51

Table 11: Supplementary latency diagnostics. Mean, p50 and p95 use completed evaluation-born messages. Mean work is the local monotonic application-processing duration. Observed violations use completion attempts occurring during evaluation and have a different denominator from the primary admitted-cohort deadline rate.

Response	Run	Decision to active (s)	Global gap (s)	Extra pending during pause	Net pending during transfer	Recovery (s)
Redistribute 3	1	27.26	15.080	10,557	7,023	180.9
Redistribute 3	2	21.92	12.307	8,614	6,019	173.4
Targeted scale 6	1	78.52	19.881	13,918	4,678	290.5
Targeted scale 6	2	81.26	21.732	15,305	5,523	266.8

Table 12: Intervention costs. Decision-to-active ends at verified ownership; transfer spans release request to verified ownership. Global gap is the longest interval without any consumer processing. Extra pending work uses that pause; net pending change uses the transfer interval. Recovery is measured from the decision to its confirmation.

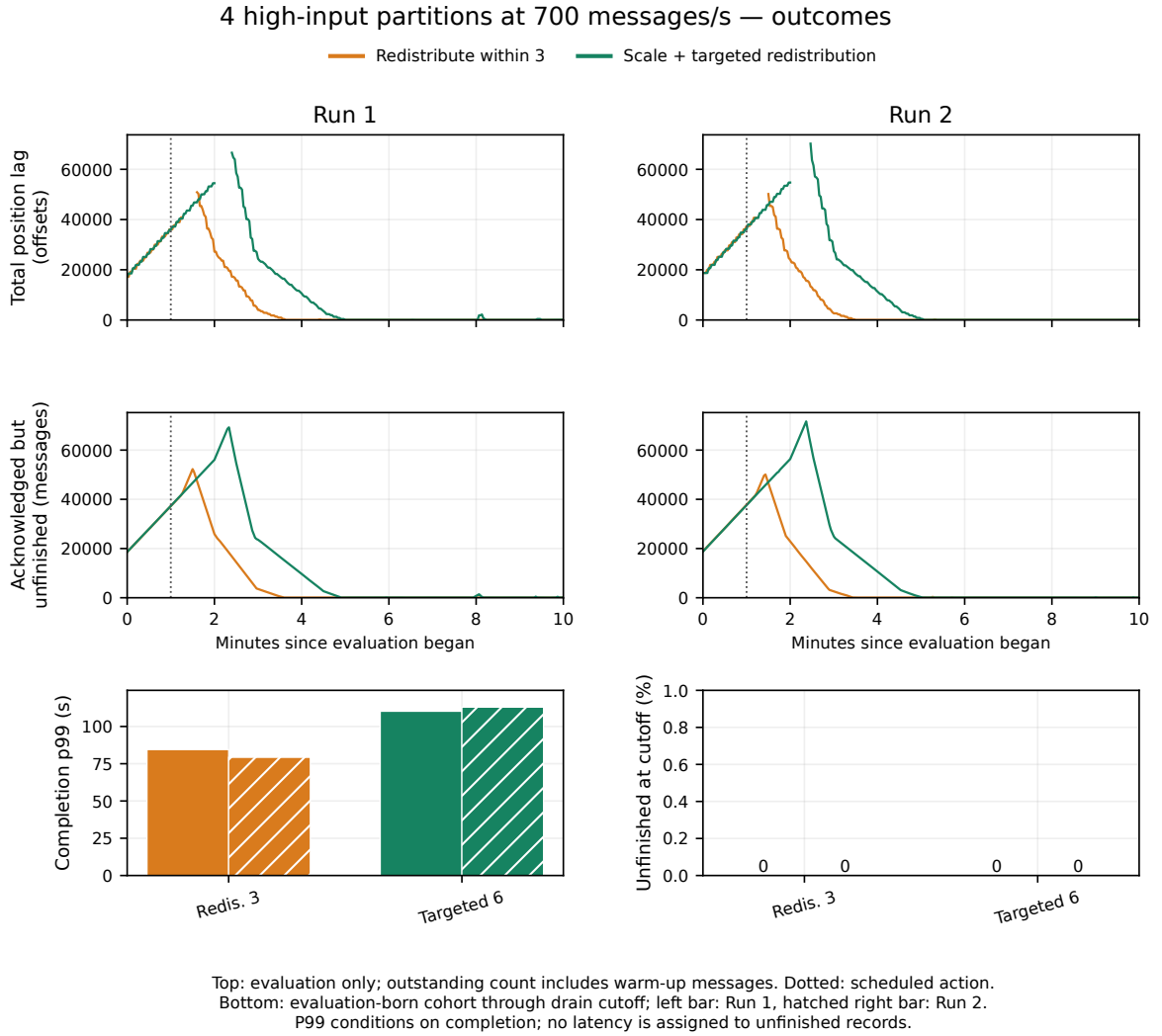


Figure 4: Compact outcome comparison: total consumer-position lag and exact acknowledged-but-unfinished work during evaluation, followed by whole-cohort p99 and unfinished percentage at the drain cutoff. Warm-up work is included in the time-series counts but excluded from the outcome cohort.

Response	Run	C0	C1	C2	C3	C4	C5
Redistribute 3	1	2	1	1	—	—	—
Redistribute 3	2	2	1	1	—	—	—
Targeted scale 6	1	1	1	1	1	0	0
Targeted scale 6	2	1	1	1	1	0	0

Table 13: Final observed numbers of configured high-input partitions per consumer. These are traffic targets, not persistent-hot detector counts. All high-input partitions initially belonged to Consumer 2. A dash means no observed partition ownership.

Response	Run	C3 (s)	C4 (s)	C5 (s)
Targeted scale 6	1	69.06	68.95	69.09
Targeted scale 6	2	71.56	71.46	71.53

Table 14: Seconds from replica request to the first valid completion on each added consumer. These intervals include startup and handover; they are not pure rebalance delays or whole-action completion times.

4 high-input partitions at 700 messages/s — performance and resources

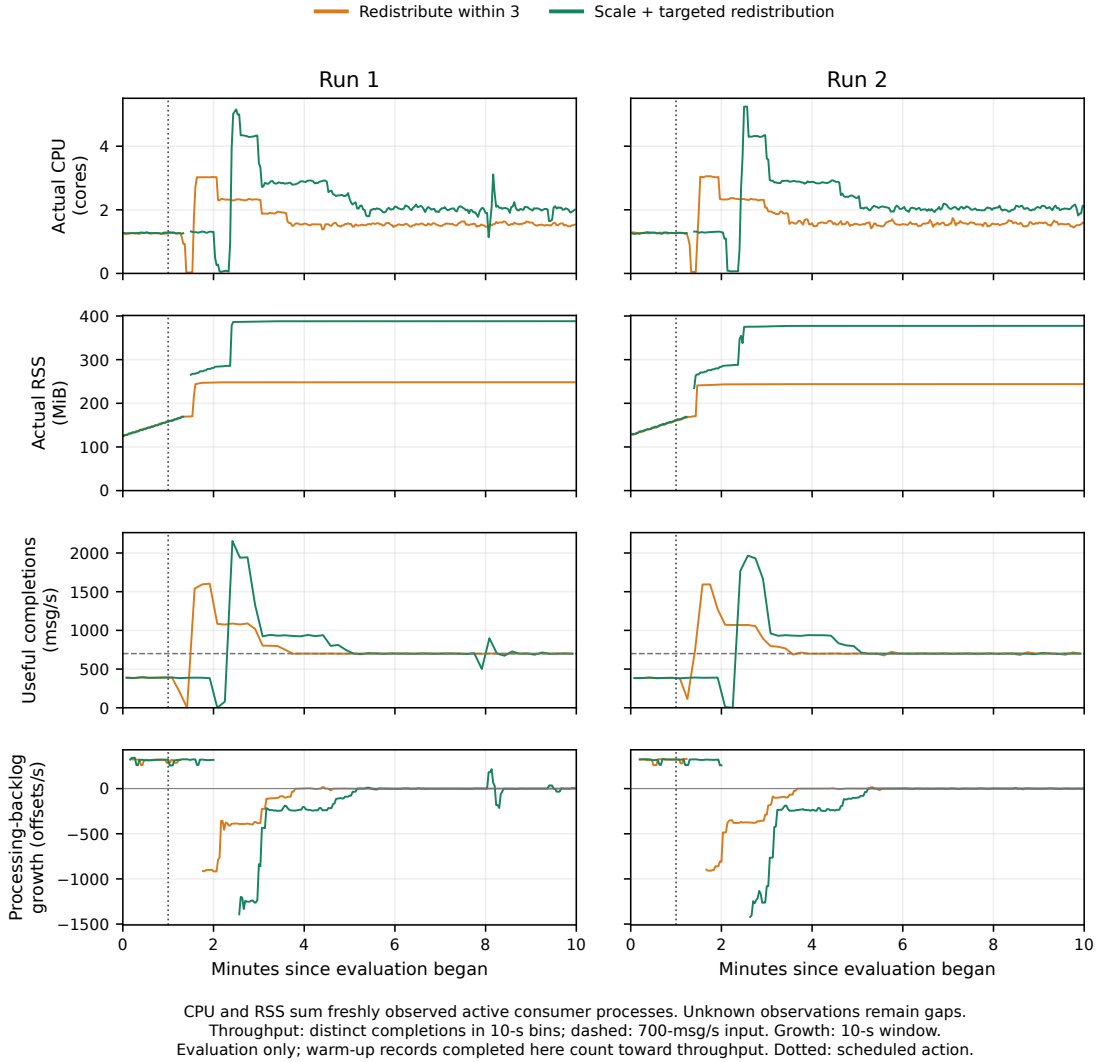
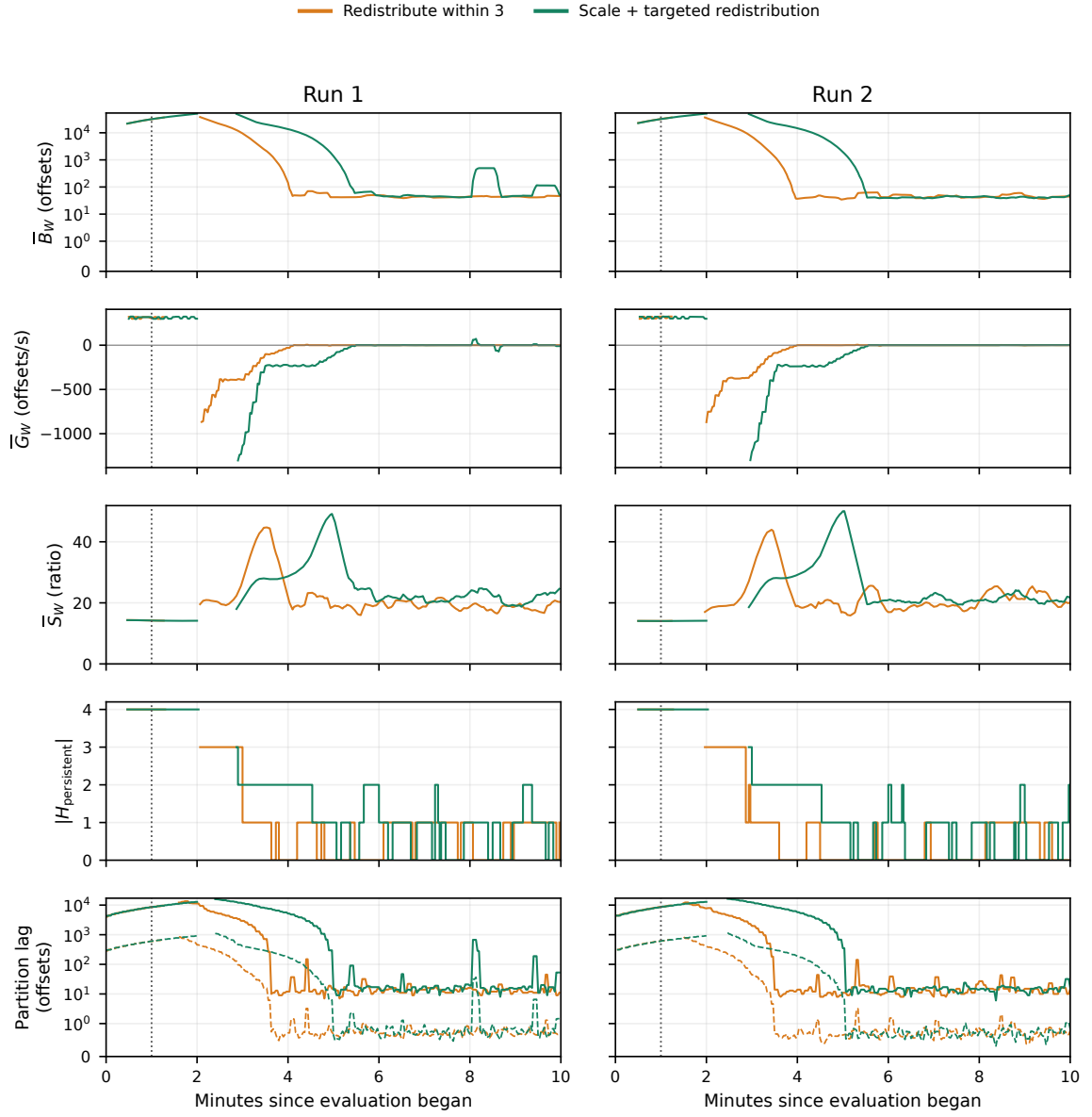


Figure 5: Actual consumer CPU and resident memory, useful completion throughput, and ten-second processing-backlog growth. Resource traces sum fresh measurements for all started consumer processes; an unavailable component leaves a gap. Requested resources and their integrals remain separate in the resource table.

4 high-input partitions at 700 messages/s — runtime state



Runtime state: first four rows; final row: maximum (solid) and mean (dashed) lag.
 B, S and persistence: 15 samples; G: 15 intervals. Dotted: scheduled action. Gaps remain missing.
 B and partition-lag axes: linear near zero, logarithmic above one offset. Hot-set IDs are retained.

Figure 6: Reconstructed runtime state and partition-lag diagnostics. The first four rows show window-mean total lag, window growth, window-mean skew and persistent-hot count. The final row shows maximum and mean partition lag. Exact persistent-hot identifiers remain in the supporting data; the count alone does not identify a reassignment target. No scheduled action was selected by this reconstructed state.

A Earlier five-minute experiments and sensitivity

A.1 Five-minute evaluation experiments: additional metrics

These two earlier blocks each used a five-minute evaluation, one-minute warm-up and two-minute drain, with 210,000 admitted evaluation messages per trial. Each block contains its own unchanged-assignment baseline and its intervention, each run twice. They remain separate comparisons rather than being pooled into the later ten-minute experiments. Their diagnostics, process resources and retrospective completion-deadline sensitivity are included here for completeness.

The historical monitoring gaps are retained. Some predate the correction that initializes a returned-record position from a verified assignment offset while the client position remains unresolved. The correction was not applied retrospectively to invent missing values. Missing observations also restart the fifteen-observation persistence history, so the diagnostic gap may last longer than the ownership transfer itself. A shorter growth window cannot recover measurements that were never available. No comparison of these blocks with the later ten-minute blocks isolates duration alone: dates, selected partition identities and the monitoring implementation also changed.

A.1.1 Five-minute evaluation: targeted scaling

Response	Run	p99 (s)	Unfinished (%)	Misses > 1 s (%)	Useful throughput (msg/s)
Keep 3	1	231.58	31.760	83.42	415.93
Targeted scale 6	1	110.70	0.000	57.01	757.32
Keep 3	2	229.28	31.037	83.24	418.90
Targeted scale 6	2	109.47	0.000	58.17	755.63

Table 15: Whole evaluation-born cohort through the fixed drain cutoff. Deadline misses include overdue unfinished messages. Useful throughput counts distinct completions during evaluation, including warm-up records completed then; it can exceed the input rate while queued work is cleared.

Response	Run	Actual CPU (core-s)	Actual RSS (GiB-min)	Requested CPU (core-min)	Requested memory (GiB-min)
Keep 3	1	515.3	1.57	42.00	21.00
Targeted scale 6	1	704.8	2.27	76.70	38.35
Keep 3	2	519.5	1.56	—	—
Targeted scale 6	2	687.6	2.25	76.81	38.40

Table 16: Consumer resources over evaluation plus drain. Actual values integrate observed process CPU and RSS intervals and sum them across consumers. Requested values integrate Kubernetes resource declarations; neither is monetary or whole-cluster cost. A dash denotes unavailable full-window request coverage, not zero cost.

Response	Run	Mean total lag	Peak total lag	Lag coverage (%)	Peak partition lag	Mean skew
Keep 3	1	58,517	101,938	99.3	8,290	4.92
Targeted scale 6	1	19,438	50,671	86.0	4,960	8.47
Keep 3	2	57,077	99,351	99.3	8,047	4.93
Targeted scale 6	2	19,473	51,721	83.3	4,245	8.92

Table 17: Lag diagnostics during evaluation only, in offsets except skew and coverage. Means use trapezoidal area divided by covered time; peaks are sampled maxima. The time mean of the skew ratio is not the ratio of time-averaged maximum and mean lag. Invalid samples and ownership changes are not bridged.

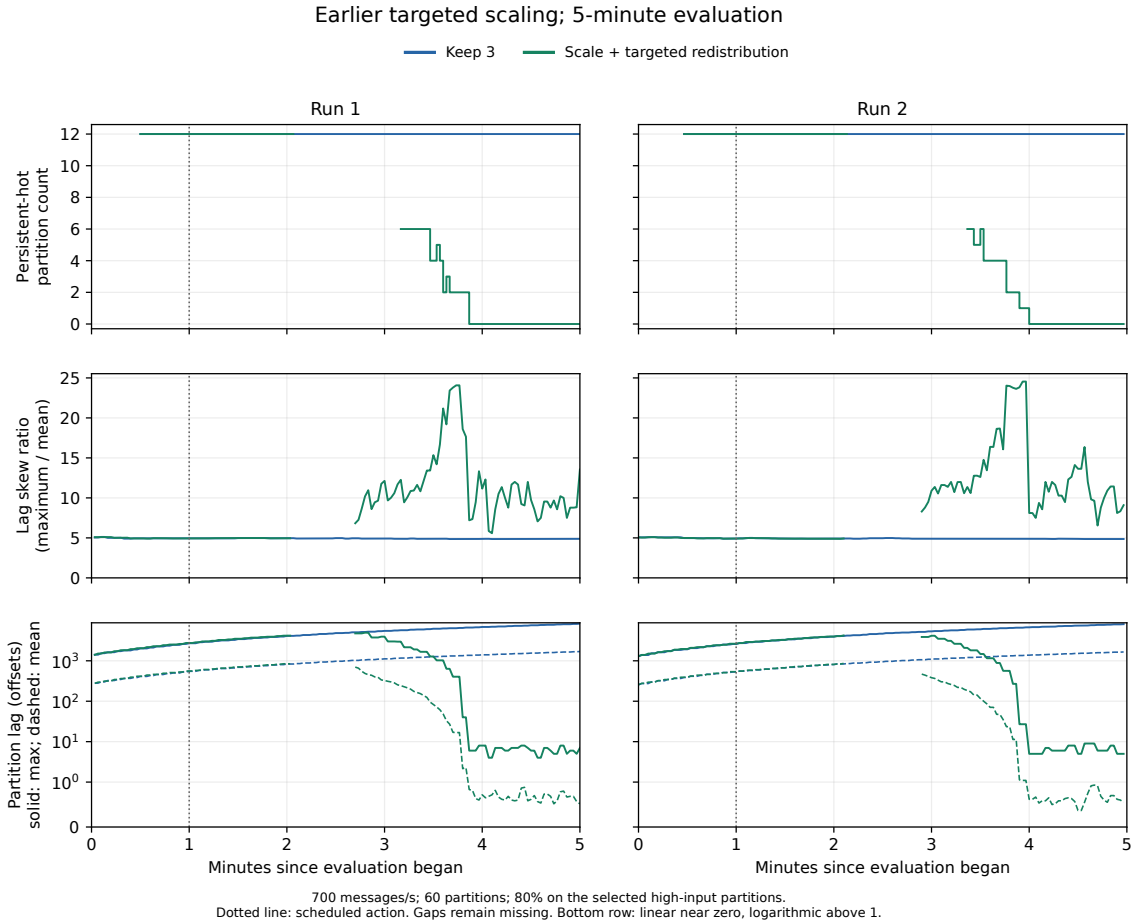


Figure 7: Earlier five-minute evaluation with the original monitoring gaps retained. Persistent-hot counts, skew and maximum/mean partition lag are interpreted together.

A.1.2 Five-minute evaluation: redistribution

Response	Run	p99 (s)	Unfinished (%)	Misses > 1 s (%)	Useful throughput (msg/s)
Keep 3	1	226.34	30.182	83.41	422.08
Redistribute 3	1	103.76	0.000	61.59	725.25
Keep 3	2	227.13	30.475	83.24	421.76
Redistribute 3	2	91.38	0.000	60.84	727.54

Table 18: Whole evaluation-born cohort through the fixed drain cutoff. Deadline misses include overdue unfinished messages. Useful throughput counts distinct completions during evaluation, including warm-up records completed then; it can exceed the input rate while queued work is cleared.

Response	Run	Actual CPU (core-s)	Actual RSS (GiB-min)	Requested CPU (core-min)	Requested memory (GiB-min)
Keep 3	1	515.8	1.56	42.00	21.00
Redistribute 3	1	554.1	1.47	42.00	21.00
Keep 3	2	516.5	1.55	42.00	21.00
Redistribute 3	2	553.4	1.46	42.00	21.00

Table 19: Consumer resources over evaluation plus drain. Actual values integrate observed process CPU and RSS intervals and sum them across consumers. Requested values integrate Kubernetes resource declarations; neither is monetary or whole-cluster cost. A dash denotes unavailable full-window request coverage, not zero cost.

Response	Run	Mean total lag	Peak total lag	Lag coverage (%)	Peak partition lag	Mean skew
Keep 3	1	58,189	99,805	99.3	8,109	4.93
Redistribute 3	1	19,243	37,159	67.3	3,724	9.41
Keep 3	2	57,329	99,153	99.3	8,025	4.92
Redistribute 3	2	17,609	35,615	68.7	3,581	9.57

Table 20: Lag diagnostics during evaluation only, in offsets except skew and coverage. Means use trapezoidal area divided by covered time; peaks are sampled maxima. The time mean of the skew ratio is not the ratio of time-averaged maximum and mean lag. Invalid samples and ownership changes are not bridged.

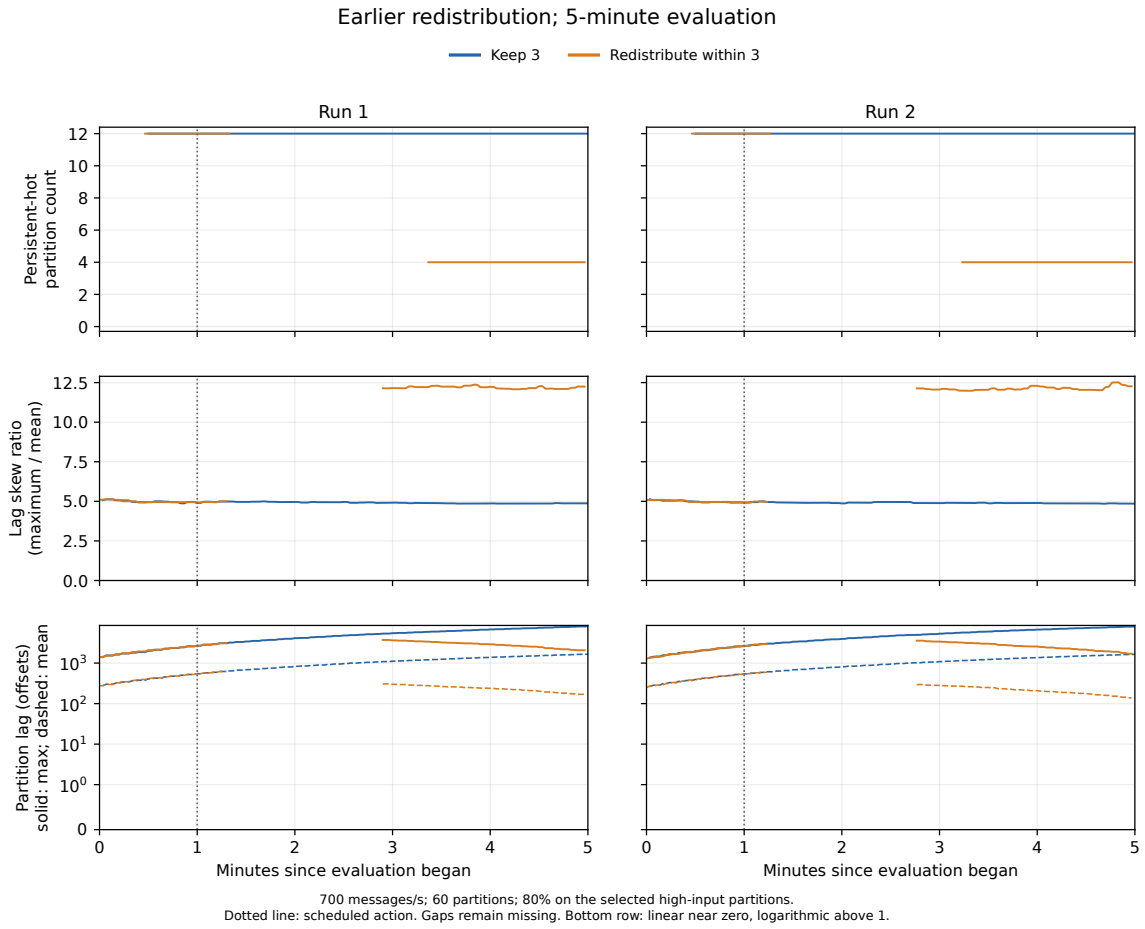


Figure 8: Earlier five-minute evaluation with the original monitoring gaps retained. Persistent-hot counts, skew and maximum/mean partition lag are interpreted together.

A.2 Completion-deadline sensitivity

The half-second deadline is retained here to assess sensitivity. The main tables use one second. An overdue unfinished record is a miss at either applicable deadline; no unfinished latency is invented.

Response	Run	Misses > 0.5 s (%)	Misses > 1 s (%)
Keep 3	1	83.336	83.321
Keep 3	2	83.388	83.364
Redistribute 3	1	30.705	30.496
Redistribute 3	2	32.200	31.838
Targeted scale 6	1	26.529	26.260
Targeted scale 6	2	27.586	27.331
Kafka scale 6	1	26.095	25.828
Kafka scale 6	2	21.255	21.051

Table 21: Twelve high-input partitions: ten-minute evaluation. Retrospective calculation; original 99-ms results remain preserved.

Response	Run	Misses > 0.5 s (%)	Misses > 1 s (%)
Redistribute 3	1	26.936	26.632
Redistribute 3	2	26.496	25.821
Targeted scale 6	1	36.053	35.232
Targeted scale 6	2	35.064	34.717

Table 22: Four high-input partitions: ten-minute evaluation. Both thresholds were prespecified.

Response	Run	Misses > 0.5 s (%)	Misses > 1 s (%)
Keep 3	1	83.447	83.418
Targeted scale 6	1	57.312	57.011
Keep 3	2	83.258	83.239
Targeted scale 6	2	58.572	58.171

Table 23: Five-minute evaluation: targeted scaling. Retrospective calculation; original 99-ms results remain preserved.

Response	Run	Misses > 0.5 s (%)	Misses > 1 s (%)
Keep 3	1	83.438	83.413
Redistribute 3	1	61.794	61.591
Keep 3	2	83.278	83.241
Redistribute 3	2	61.100	60.840

Table 24: Five-minute evaluation: redistribution. Retrospective calculation; original 99-ms results remain preserved.

B Measurement Coverage

Table 25: Summary of the defined metrics, where their results are reported, and measurement limitations or remaining planned evaluations.

Metric family	Evidence in results	Interpretation
Partition lag and total lag	Main lag plots, diagnostic figure and lag-summary tables; older five-minute counterparts in the appendix.	Offsets, not guaranteed record counts.
Processing backlog and growth	Main backlog-growth plots and acknowledged-unfinished plots; growth is also retained for consumer-position lag.	Completion frontier and returned-record position have different definitions.
Maximum and mean partition lag	Bottom row of each diagnostic figure and sampled-peak tables.	Maximum and mean are taken across partitions at each valid observation.
Lag skew and window-mean skew	Runtime-state row shows window-mean skew; the summary table and separate diagnostic plot retain raw skew.	High relative skew can coexist with small absolute lag.
Hot set, persistence score and persistent-hot set	Persistent-hot row, detector settings and interpretation of early versus late observations.	Exploratory diagnostic, not a calibrated controller trigger.
Window-mean backlog and runtime state	Runtime-state figure and timestamped CSV/JSON, including persistent-hot partition IDs.	Reconstructed from retained observations; no adaptive controller was evaluated.
Completion mean, p50, p95 and p99	Outcome and latency-detail tables for each block.	Each quantile uses completed evaluation-born messages, not averaged rolling quantiles.
Processing-start latency and processing duration	Latency-detail tables and separation of queued delay from synthetic work.	Processing duration excludes evidence writing and commit bookkeeping.
Unfinished work and completion deadlines	Unfinished percentage and 1-second cohort deadline misses in main tables; 0.5-second sensitivity in appendix.	Overdue unfinished messages count as misses; they have no fabricated completion latency.
Observed-completion violation rate	Separate latency-detail column at 1 second.	Completion-attempt population during evaluation differs from the admitted cohort.
Input, useful throughput and attempt throughput	Main throughput plots and outcome tables; admitted rate and replay checks in text.	Attempts and distinct completions agree only when no duplicate completions occur.
Actual CPU and resident memory	Main process traces and observed resource integrals.	Consumer processes only; no full-cluster energy or monetary-cost claim.
Requested CPU and memory cost	Main resource tables, integrated over evaluation plus drain; detailed resource-cost figures retained with records.	Partial request coverage is unavailable for full-run comparison, not zero.
Interruption, transition backlog and recovery	Main intervention-cost tables, with operational endpoints identified.	Recovery is first threshold confirmation while input continues; not permanent stability.
Correctness and measurement quality	Identity reconciliation, handover validation, clock limitations and explicit coverage discussion.	Synthetic records do not prove exactly-once business effects at a durable external sink.
Controller overhead, splitting and state-transfer cost	Explicitly identified as unevaluated future measurements.	No adaptive controller, hot-key splitting or stateful transfer was evaluated in these trials.